

VR Environment of Digital Design Laboratory: A Usability Study

Mariam Ali Alnuaimi

Department of Computer Science and
Software Engineering, UAE University
Email: 201600912@uaeu.ac.ae

Mamoun Awad

Department of Computer Science and
Software Engineering, UAE University
Email: mamoun.awad@uaeu.ac.ae
ORCID 0000-0001-7424-4129

Abstract

Virtual Reality (VR) offers an immersive learning environment with the potential to revolutionize how students engage with complex topics and interact during classes. Great care must be taken when designing and developing VR applications. This paper presents a usability study of our proposed educational VR application designed for the Digital Design & Computer Organization lab. The VR Lab is meant to support the traditional lab that is taught in college first-year. To assess the VR application, we conducted a usability study with a wide range of students from different colleges, evaluating their performance on tasks within the VR environment and collecting their feedback on the application's design, functionality, and ease of usage. By analyzing quantitative data (task completion rates, time spent) and qualitative feedback (surveys, observations), this study identified potential usability issues and recommended improvements to enhance the learning experience within the VR application. The usability study showed that the majority of students found that the VR application immersive, engaging, and viewed their experience as helpful to better understand complex concepts. Additionally, the study found that careful design and technology must be used selectively due to population variation in terms of technology knowledge level, health factor, and readiness.

Keywords

Virtual Reality (VR), Digital Design & Computer Organization, Usability Analysis.

1. Introduction

The fields of engineering and computer science education are constantly seeking innovative methods to engage students and deepen their understanding of complex concepts. Traditional laboratory environments, while valuable, can sometimes lack the ability to fully immerse students in the subject matter. In many cases, engineering and computer science students spend between 50-100 hours a week on their course work, including lectures, labs, assignment, and projects. Typically, most of the physical time is spent in the lab because projects and assignments are conducted in the lab. Studies showed that students suffer from frustration, fatigue, and tiredness due to hardware failure and/or advanced concept confusion [15]. Virtual Reality (VR) can provide an immersive, competitive, and fun experience during lab classes which can be considered an effective solution to lab limitations. In addition, the use of VR in engineering pedagogy has been adopted widely [22], and it has been shown that VR-based laboratories are very supportive and effective in engineering, though it cannot replace traditional in-person laboratory experience [24]. This is due to the nature of the labs in engineering, which requires dealing with wires, connections, leaderboard, etc. Furthermore, VR provides effective accessibility to lab virtual resources because in many cases small glitches and hardware failure might delay the student and cause frustration waiting for lab assistant help.

VR can overcome such limitation because VR application will be available all the time with supportive and relevant virtual resources. Moreover, there is an additional cost for VR sensory equipment, though prices are getting affordable for both students and institutions.

VR in education has several advantages that include inclusiveness, visual quality, interactivity, cost-effectiveness, and safety. On the other hand, VR users can suffer from psychic risk and cyber sickness. Also, VR technology is complex and has a high cost [1][14]. Additionally, VR applications require thorough usability testing to guarantee the best user experience and satisfaction [17][18]. Therefore, VR applications must be designed and developed with great care, as well as it must be thoroughly validated and tested. Usability testing can detect any usability issues in early stages, evaluate immersion and presence, identify interaction issues, such as voice commands, gaze, gestures, etc., and reduce motion sickness and discomfort.

Unfortunately, the teaching methods in Engineering and Computer Science are still traditional and little attempts have been proposed seriously on using VR environments on large scale. In addition, many VR applications that were developed for research studies are simple, limited, and lacking usability testing. We believe that there remains a dearth of research developing complex VR applications that comprise different educational features, as well as integrating usability studies in those applications. In this paper, we attempt to transform a digital design lab into VR environment to explore the usability, suitability, and learning experience of students. Specifically, we investigate the usability of a VR application specifically designed for a Digital Design & Computer Organization lab. By creating an interactive and immersive learning environment, VR applications can revolutionize how students engage with lab topics. To assess this potential, we developed a VR application to educate students on digital design and computer architecture circuits. We conducted a usability study with student volunteers. The study evaluated user performance on assigned tasks within the VR environment and gathered their feedback on the application's design, functionality, and overall ease of usage.

Using standard evaluation metrics and through a comprehensive analysis of both quantitative data, such as task completion rates, and time spent, and qualitative feedback, such as surveys, and observations, this study aims to achieve two key objectives. Firstly, to identify any usability issues that may hinder student learning within the VR application. Secondly, through these findings we would recommend specific design changes and improvements, ultimately enhancing the educational value and user experience of the Digital Design & Computer Organization lab VR application. The main contributions of this work are summarized as follows.

1. We developed VR application to support student experience in Digital Design and Computer Organization course.
2. We conducted surveys and a usability study to compare between learner's task performance and perceptions with VR environment vs. traditional learning environment.
3. We investigated and analyzed the usability study results and suggested solutions to the issues found while using the VR application in engineering.
4. We augmented and measured the use of gamification aspects to the VR application to test the effect of gamification on students understanding.

The remainder of this paper is organized as follows: Section 2 summarizes related and relevant research work. Section 3 presents the research methodology including research questions, VR prototype, and usability metrics used in this paper. Section 4 elaborates on the usability study experiments and data analysis. In section 5 we discuss our usability study findings. Sections 6 and 7 highlight the limitations for this study and the ethical considerations. Finally, in Section 8, we draw our conclusions and discuss future work.

2. Literature Review

VR is a powerful tool that is used vastly in games, simulation, and most importantly training and education. The literature is very rich in research on different dimensions in VR learning. In this section, we present abroad comprehensive review on VR learning. First, we present various research work/reviews in VR in education, second, we elaborate on the core effective factors that govern VR learning and assessment, and finally, we highlight relevant VR research work on usability studies.

2.1 VR in Education

The use and integration of VR in education has several benefits and proved very effective in enhancing collaborative learning [26] and improve students' comprehension [2][27]. In addition, VR provides excellent immersive and engaging learning experience [25]. Furthermore, in high-risk environment with complex machinery or hazardous materials, VR creates a secure and regulated learning environment for students [29]. For example, in medical training [31][32], disaster preparedness [33], safety and cost-effective training [34], trainees can repeatedly practice hazardous experiments or emergency procedures without real-world risks. Moreover, VR can drive cost down [28] because we substitute hardware with virtual environment that can be accessed by many students simultaneously and allows for repeated sessions [39][40]. Also, VR proved to enrich students' comprehension of different engineering subjects due to its ability to connect theoretical complicated concepts with practical visualized and animated applications [30]. As such, students acquire potent technical skills and gain soft skills and confidence while facing future challenges after graduation. Besides, VR features accessibility and inclusivity in which remote students or students with disability can enjoy access to lab environment or field experiences [35][36]. Finally, learners can get real-time feedback and instant assessment and guidance [37][38]. Soliman *et al.* [19] surveyed different research in engineering education and presented evidences that VR is an excellent tool to improve students' performance and experience. Similarly, the surveys in [20][21][23] reaffirmed the increasing popularity and impact of VR in education. However, they found that more extensive exploration in engineering education need to be conducted with clear evaluation metrics in realistic VR scenarios. Feisel *et al.* [24] found that VR-based laboratories can provide alternative laboratory experiences in engineering without automatically replacing the traditional in-person laboratory experience. One clear support of VR platforms in education is that hands-on activities and the acquisition of procedural skills can be rendered and repeated in VRLE-based training systems [25].

On the other hand, VR technology has health issues. For example, goggles, headset, tracking sensors have shown to cause users to suffer from psychic risk, eye strain, cyber sickness, or disorientation [5]. Additionally, VR might have high initial cost and requires expensive software and hardware maintenance [39][40]. Moreover, VR can pose technical barriers for some users due to usability challenges. Also, user might face Internet accessibility problems such as network latency issues, bandwidth limitation, and connectivity disruptions [35]. Additionally, the development of VR applications requires substantial resources and real-life sensory experience which dictates thorough usability testing to guarantee the best user experience and satisfaction [17][18][25]. Finally, it was shown that VR can negatively affect learners' isolation, communication skills, and social interaction [41]. Nonetheless, with the rapid advancement of VR technology, the integration of VR in academic education system becomes just a matter of time [30].

2.2 The Effective Factors in VR Learning

The use of VR in learning is governed and assessed using different factors that include several factor such as realism, contextual relevance, context, coherence, immersion, and presence [47]. In the following, we will explore the current research on the key factors of effective VR-based learning.

Many researchers consider VR as another kind of fiction and argued that it should be conceived of as an experience rather than a technique, and such experience might gain more from increased emotional relevance than from higher realistic resolutions [45]. We should focus on human aspects, activities, and features in the VR system that transform user experience to reality. The transfer of the user experience from the design of VR interactive systems to the real world is a key success factor [59].

Realism is the degree of VR system to mimic the real world, and it is an important factor in VR learning. The authors in [46] conducted two studies to examine how realism of environmental presentations impact affective responses and participant perceptions. They found that the level of realism acquired with VR does impact affective responses and perceptions. In addition, most of the studies reported that higher realism has a positive impact on user experience [49].

Presence and Immersion are two core factors in VR learning. Presence refers to the qualia of existing in a real place. Immersion, according to Slater [50], refers to the technology giving an inclusive, extensive, surrounding, and vivid experience to users such that the user loses awareness of the fact that she is actually in an artificial world. Katernyak *et al.* [51] studied the cognitive presence and the effect of immersion in Virtual Learning Environment. They considered each learning task as an event where participants might have enjoyed memorable experience. They conducted cognitive experiment to learn the percentage of students who get immersed memorable experience when involved in different eLearning activities. In their cognitive experiment, 328 students participated in an e-course on project management. The participants were challenged with a quiz on remembering and understanding, and a task on understanding and application in an interactive context. Results of the survey and interview uncovered that 41% of the e-course participants had the feeling of arousal and 'drive' (awakening, excitement), while 23% of the participants felt freedom, joy, sense of satisfaction, influx of new ideas. The authors concluded that the success of e-courses is defined by the following factors: objective of the e-course, monitoring the progress of the learning process, the social presence, and cognitive presence which encourages collaboration, creativity, and idea generation. Selzer *et al.* [52] studied the relationship between virtual presence and the learning outcome arising from a VR educational experience. In their experiment, 42 participants were randomly assigned to one of three experimental groups: desktop computer, low-end VR configuration, and high-end VR configuration. In their results, they reported a lower learning outcome in the less immersive configuration (i.e., desktop) and a similar learning outcome in both low-end and high-end VR systems. They argued that even though low-end VR systems are less immersive and produce a lower level of virtual presence than high-end VR systems, the results support the use of low-end VR systems for educative applications.

The above-mentioned core factors correlate in their effect on VR learning. In their review on the correlation between coherence, realism, illusion, and immersion in VR, Jung *et al.* [47] proposed two new models for representing the correlation between coherence, immersion, and illusion. The first model has three orthogonal axes representing coherence, immersion, and illusion. The axes describe the subjective feelings that arise when a user lives VR experiences. They found that each factor can be treated as independent dimension, but that they might partially influence each other, as they intersect. In order to measure lower-level components, they proposed the second model. They hypothesized that increased realism has positive impact on deep presence, including co-presence and social presence. The study did not include or conduct any experiment to validate their two models. The work in [48] attempted to identify features and effective factors which may contribute to learning environment in Desktop VR design. The authors applied content analysis which integrates meta-analysis to achieve their goal. The proposed a framework that is comprised from instructional element, interaction, navigation, and fidelity. The authors claimed that their model can empower the cooperation and understanding in all areas of learning, including VR environment, though no validation or testing was conducted on the framework.

VR environment and tools enhance students' engagement and interactivity in visualization learning. The authors in [53] developed a 3D visualization tool for engineering students. They conducted a cognitive experiment on 20 undergraduate students. Their results showed that VR application is very effective on users' exploratory and analytical activities. Their statistical analysis indicated a significant improvement in both the post-experiment test scores and the engagement level, which was measured via a user engagement questionnaire and participants' comments. In another similar experimental study on interactivity levels in VR learning environments [54], the authors found insignificant difference in students' learning using pre and post tests, however, questionnaire and interview data suggested that interactivity in the experience significantly affects students' engagement and focus in learning.

In this paper, we measure the engagement and interactivity of the students in a VR-Lab on computer architecture for computer science and computer engineering students. We use survey and interviews to assess students' experience and investigate the effect of usability study on VR development in education. In addition, we explore the effect of VR-Lab on engineering education and engagement to decide and recommend future decision on the integration of virtual lab in the curricula.

2.3 Usability Testing in VR Learning

Usability testing in VR applications plays a crucial role in gathering data needed to ensure the effectiveness of the application and identification of the application issues. Usability testing is used to measure the VR application's learnability, efficiency, memorability, errors, and satisfaction which allows the developer to get direct feedback from the user [3]. There are 2 types of usability tests: formative and summative user testing. The formative test is used earlier in the testing process. It aims to identify interface design problems and provide solutions. Observation testing of the quality of the application is conducted during the formative test on a small group of users [4]. The summative test is used to evaluate the efficacy of the final application design. Typically, it is conducted after the launch of the application and on a larger group of users. The aim is to collect large amounts of data to measure the performance of the users in using the application [3].

With the rapid development and accessibility of VR applications in different areas, the need for objective data-driven usability testing is becoming important venue of research that needs to be explored, because traditional methods of testing are both time and resource consuming, and in many cases induce highly subjective results. The authors of [3] explored the venue of automation of usability testing of VR applications. They considered objectives features such as HMD built-in head and hands tracking, EEG sensor, video recording, and other measurable parameters in addition to automated analysis of subjective data provided in questionnaires. They conducted experiments on 14 volunteers who tried a VR application with relatively easy tasks that did not generate stress for the users. In addition to record the participants signals, the observer recorded the participant behavior during the experiments and collected post-experiment feedback. The analysis of the results showed promising conclusion that automated usability testing of VR applications is potentially achievable.

In a recent usability study on VR environment [18], Huang *et al.* showed that that there are three major usability factors in 3D modeling learning in VR work: interactive quality, dynamic compatibility, and flow effects. Zilles Borba *et al.* [58] demonstrated in a control experiment study that the application of systematic usability methods during cyber system interactions is very effective. Their results showed how the usability standard of ISO 9241-11 can be used for a set of efficiency, efficacy and satisfaction measures. They also analyzed the level of presence in the immersive virtual environment and issues related to cyber-sickness.

The authors of [5] showed that in VR environment immersion must be considered in usability testing rather than traditional Human Computer Interaction (HCI) methods which are not entirely applicable. There are three types of immersion: full immersion, semi-immersion, and no immersion. VR systems that are fully immersive use head-mounted displays (VR headsets) such as Oculus Quest, or HTC VIVE combined with haptic devices. A semi-immersive VR environment consists of large displays in which the content of the VR application is projected for the user to look at. This could limit the interactions between the user and the environment. A non-immersive VR system is when the VR environment is displayed on a computer monitor [5]. Additionally, the authors found that when conducting usability testing on VR environment, we should also consider other issues such as health and safety. For example, immersive VR requires the user to be fully involved and when the VR equipment is not used correctly by the users it may lead to discomfort and damage. Some of the health and safety considerations include eye strain because of the HMD displays, claustrophobia, injury or trauma due to unprepared testing environments or sudden movement, and cybersickness which is the equivalent of motion sickness. The user must be warned about health concerns before starting the test. Brade *et al.* [6] compared between the virtual field environment and the real field environment. User experience was a key factor in their usability test. They compared how the users would navigate a real-world environment to a virtual environment. The test contained about 60 participants. Through this test, the study found that the users preferred the virtual environment when it was more engaging. This suggests that a well-designed VR environment could potentially replace a real-world environment. Locus J. [7] developed VR environment to investigate the potential of VR to improve student's understanding of construction assemblies. The study uses a Head-mounted display and an X-box controller for the student to experience this VR environment. To test the simulation, the study used pre- and post-survey techniques in which the test sample was divided into two groups: (a) prior video game use, in which the students are given images and traditional lecture material to study the course material, and then they take the survey, and (b) Prior HMD-VR use in which the students were given a pre-survey on the same lecture material and then they were given the chance to use the VR environment. After the VR experience, users were given a post-survey and asked about usability and comfort. The study concluded that the students who used VR had a slight increase in understanding the materials. According to [8], VR environment allows users to feel like they are inside the design, potentially leading to a more comprehensive understanding of its form and function. Additionally, VR could facilitate easier interaction with the model, allowing the users to manipulate it and identify potential problems more intuitively. Moreover, the study suggests that VR design reviews might lead to fewer missed design flaws compared to traditional methods. The immersive environment could foster better communication and collaboration among reviewers due to a shared understanding of the design in a 3D space.

The work in [7] uses an Xbox controller which can be used by pressing buttons and moving the joystick. The controllers we use in our project are sensitive to movement meaning that the user will see his hands move in the application just like real life. The buttons in the back of the controller give the user the feeling of grabbing the actual objects in the application. de França *et al.* [55] argued that the design and development of VR applications that suit all kind of users pose a serious challenge. Therefore, it is necessary to investigate the usability of these technologies as well as the ergonomic constraints and risks inherent in both the physical devices and the characteristics of the virtual environment adopted by the user. The authors proposed to observe both verbal and non-verbal behaviors. As such, the ergonomist might need to provide specialized equipment for the recording/monitoring of such behaviors. Ultimately, the results will provide support for any ergonomic improvement including system redesign. In the study conducted in [56], Chen *et al.* proposed a modified group usability testing to minimize data loss and to collect more significant critical data with lesser time, cost, and effort. They developed a non-immersive VR environment and conducted experiment on a sample of 36 participants to compare between the group usability testing vs. the

modified one. Typically, group usability testing involves many-to-many participants executing tasks simultaneously, with one-to-many observers interacting with the participants to generate abundant useful data in short period of time [57]. In the proposed modified approach, observers record their observations and participants briefly note down any usability issues encountered during the testing session. In addition, the observers request from the participants to actively verbalize the noted usability issues or problems. In their experiments, 4 level tasks and different comparison criteria were measured. The results indicate that the modified group usability testing is more effective and efficient than the original approach as it can collect more critical and significant data with lesser time, cost and effort consumption.

The work presented in this paper is similar to the work in [5][6][7] since they discuss the importance of usability testing and promote VR integration in different environments. However, our work focuses on the use of VR in the context of education and how it can improve students' understanding in a scientific topic. In addition, we accentuate the use of gamification and immersion in VR. We argue that adding gamification aspects such as levels, hints, score, lives and leaderboard can improve engagement and user experience.

3. Research Methodology

The research methodology in this study employs mixed-methods approach, combining quantitative and qualitative data collection techniques to gain a comprehensive understanding of usability attributes and user experience within the Digital Design & Computer Organization lab VR application. Additionally, we use surveys to gather user feedback, comments, preferences, and experiences.

Figure 1 presents our research methodology which includes defining research questions and goals, developing and implementing VR prototype, designing usability quantitative metrics, developing surveys, and finally collecting data and analyzing the usability of the VR system. The following sections present each step and provide elaborate explanations.

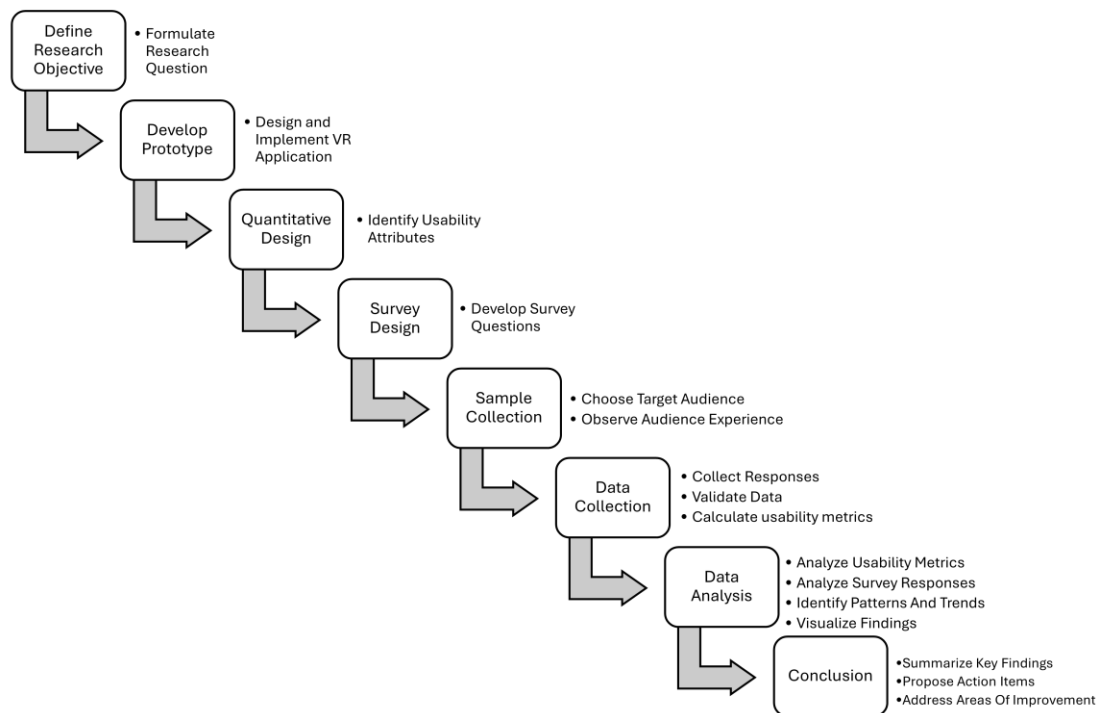


Figure 1 Research Methodology Steps

3.1 Research Questions

This study aims to identify the extent to which our VR application of digital design and computer organization Lab can be used by a specified audience and achieve the goal of replacing the physical environment, composed of breadboard, when needed with effective, efficient, and easy-of-use VR application. It also aims to identify any design issues in the VR application and find solutions to these issues. Based on that we define our research questions as follows.

- Which tasks the users struggle with and take longer time than others?
- How long does it take users, on average, to complete different tasks within the VR environment?
- What types of errors do users encounter while performing tasks in the VR environment (e.g., navigation errors, data entry errors)?
- How satisfied are users with the overall usability and functionality of the VR application?
- Do users find the VR gamified environment engaging and helpful for learning the presented concepts?

3.2 Prototype Development

Digital Design and Computer Organization (CENG205) is a core undergraduate course in the College of IT at UAE (United Arab Emirates) University. The course introduces students to the basic concepts of digital systems, digital logic gates, truth tables, and combinational and sequential logic circuits. The course is equipped with hands-on labs that bridge the gap between theory and practice. In this section, we present the specifications of the VR application of digital design & computer organization lab. The VR application simulates the physical Lab of CENG205 and includes hands-on the five essential logic gates, namely, AND, OR, NOT, NAND, and XOR. Additionally, it includes two complicated level gates: 1bit comparator and half adder. This VR environment will enable us to observe students' progress in learning logic gates. Moreover, the VR application supports gamification techniques to enhance students' engagement. The main purpose of the prototype is to allow students to learn and conduct the experiments independently.

Figure 2 presents the use-case diagram of our VR application. The use case diagram is a UML standard to model functional requirements. The VR prototype provides different functions to the actors/users such as login/register, viewing lab materials, conducting lab experiments in different levels, showing scores, redoing lab experiments, and others.

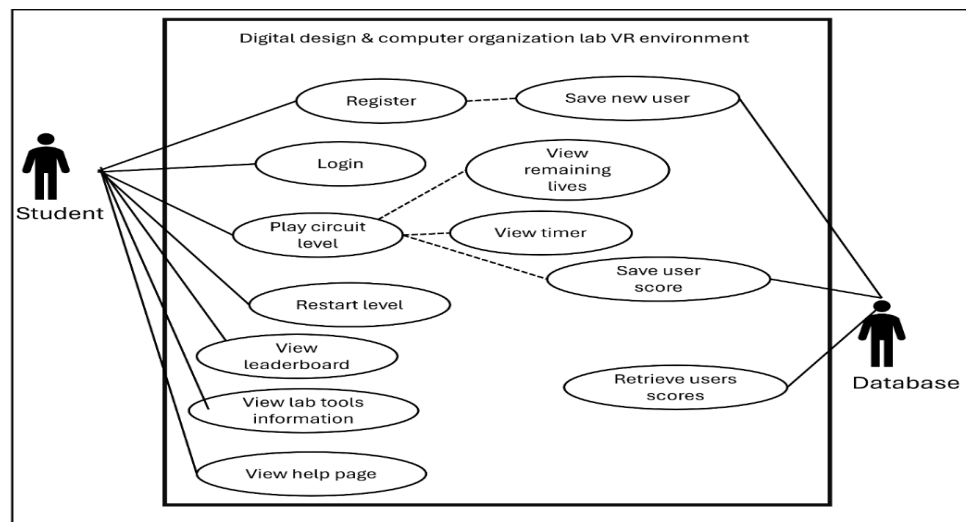


Figure 2 Use-case diagram of the VR application

Figure 3 summarizes task workflow of different features and screens that the student encounters while navigating in the application. The application is designed to help student easily navigate and conduct labs to gain knowledge. It is equipped with ROBO gadget that provides written and audio assistant to students to guarantee full scale benefits.

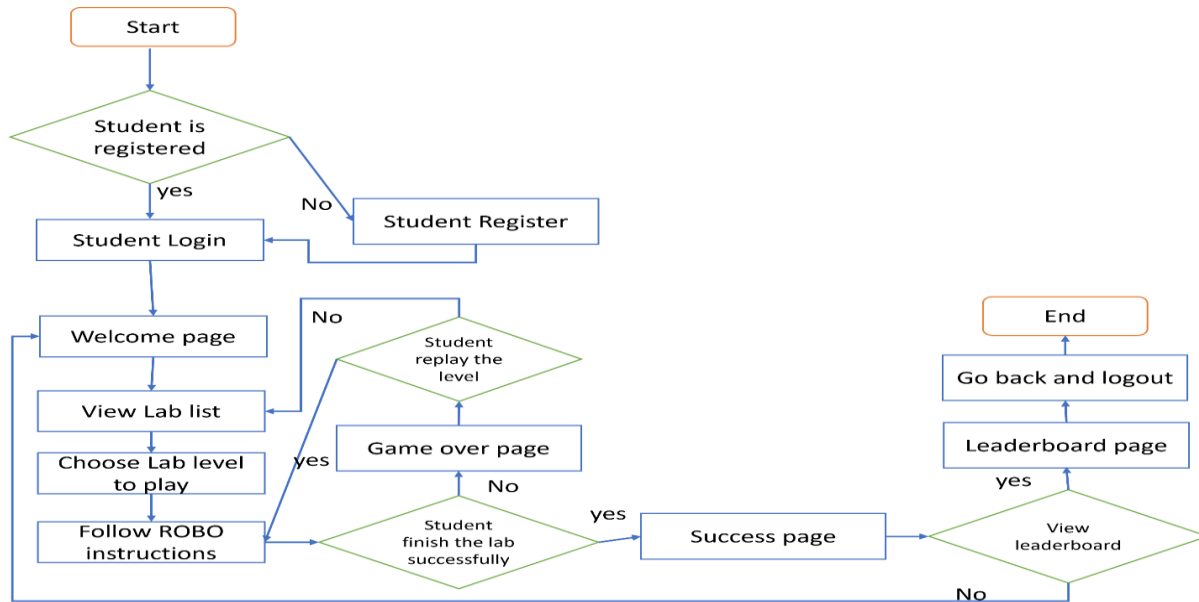


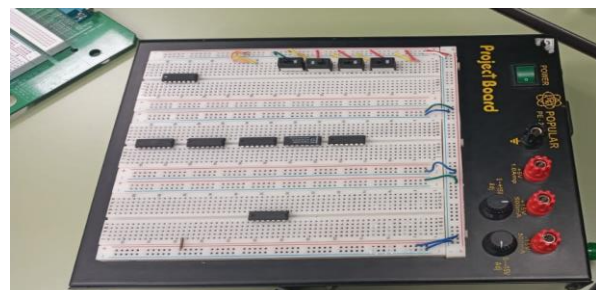
Figure 3 VR application task workflow.

In the following, we present briefly the main functions and screens of our gamified VR educational application. More details on software requirements specifications (SRS), use case specifications, and UI can be found on GitHub [16].

Figure 4 presents the physical Lab environment. Typically, the lab has chairs and desks (A), and physical leaderboards and wires/gates (B). The lab instructor distributes the lab exercise to the students with preset goal, steps, drawing, and hints, see Figure 5. The students spend 1 hour per lab to wire the gates on the breadboard. The lab instructor provides help, hints, and trouble shooting. As we can see, the time, space, and resources are limited/restricted during the lab, which might result in frustration and lack of understanding.



(A) Physical Lab Setup



(B) Physical Breadboard

Figure 4 The physical Lab setup and environment

Experiment 4 – Arithmetic Applications

[Course Outcome: 3,6]

Objectives

The purpose of this experiment is to study the internal structure of binary adders and to test the functions of these adders in a practical circuits. Students will use basic digital ICs to build circuits

Half Adder:

Half adder cell is a circuit that adds two binary numbers, one bit each and produces two outputs, S for the sum and C for the carry, together the S and C they form the result of the addition. HA schematic and its truth table is given below, you are to fill up the truth table then build the circuit using 7408 (4 AND gates) and 7486 (4XOR gates)

Input		Outputs	
A	B	C	S
0	0		
0	1		
1	0		
1	1		

Full Adder:

A full adder cell is an expansion of HA, it adds 3 binary numbers, one bit each, the differences is that FA considers the carry in that may be generated from previous addition operation and sum it with both inputs, at the output we have again two outputs, S for sum and Co for carry out. The schematic for FA cell is given below so you are to fill up the truth table provided and writ the logic expression for each of the outputs. Once done, use your proper lab equipment and build up FA cell and verify its functionality along with your truth table

Input			Output	
Ci	B	A	Co	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

You must present your circuit to your lab instructor at the beginning of the next lab session

Figure 5 Sample Lab Text

In comparison with the physical lab setup and hands on, we will present the VR Lab in the following figures. Figure 6 shows the main landing page which includes application title, login, and register options. The user can login or register through dedicated pages, see Figure 7, so that the application can keep track of students' scores and progress. Figure 8 presents the available educational labs that the student can view and provides options to start/stop the Lab. The figure also presents the leader-board option.



Figure 6: Landing Page (Main menu): The player chooses the Login or Register option.

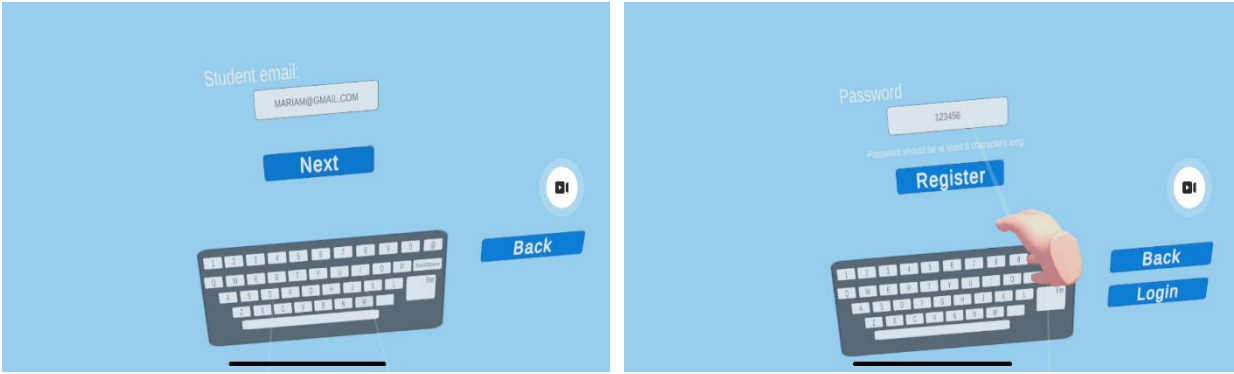


Figure 7: Register page: the player is entering her email and password.

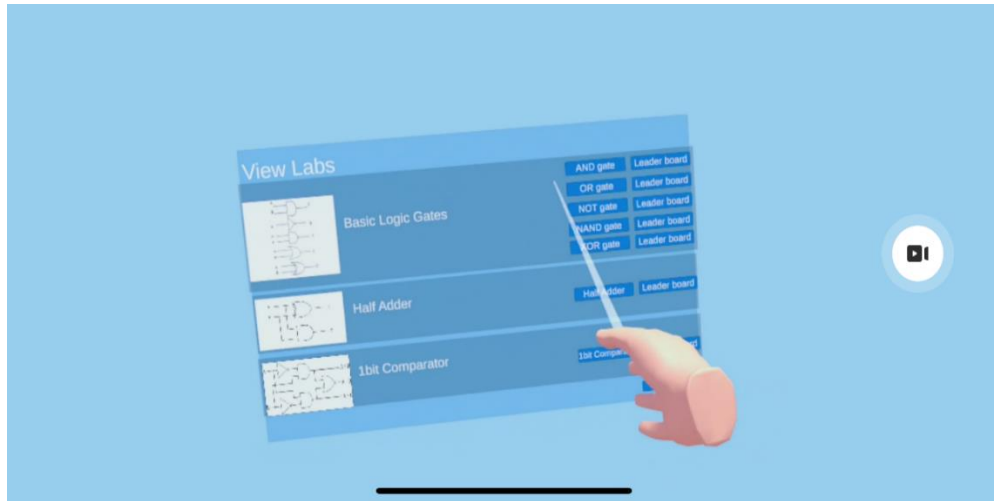


Figure 8: Lab list page, where the student can choose Lab to learn

The student can conduct the lab by following steps recited by a virtual robot. Figure 8 presents different lab levels in which the student can try and learn. The Lab page includes a virtual robot (ROBO) companion that guides the student through the lab with voiced instructions, Figure 9, and written instructions on the breadboard, Figure 10. While conducting the lab, students will be able to use the controllers as hands to carry objects and place them in highlighted areas as guided by the ROBO. They will be able to see hearts on the left side. The hearts represent the attempts to submit the steps. Each lab will provide 3 hearts in which the student is redirected to the game over page if they skipped a step or did not follow the instructions correctly three times. On the right corner of their vision, they will see a timer that starts as soon as the lab starts. The timer will record each student's time score and save them on a leaderboard. The student can exit and leave the lab, see Figure 11.



Figure 9 ROBO is giving instruction to the student

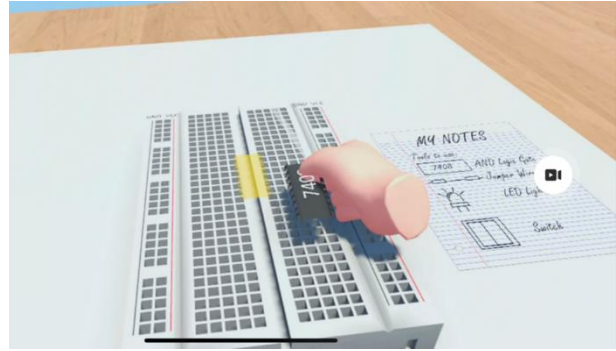


Figure 10 Hints are shown on the breadboard



Figure 11: Conducting Lab level, 1-bit comparator with truth table and K-maps hints

The student can check her score and progress through the leaderboard page, Figure 12. Each student is displayed on the leaderboard showing her score and total time spent to complete the lab, i.e., the less time the student spends on the labs, the higher her placement is, see Figure 12. The student can get more documentation and help to read and learn through the help page, see Figure 13. Notice that the VR-Lab environment provides online help, documentation, hints, and unlimited resources. In addition, the student has no time or space limitations and can access the resources and repeat the lab many times.



Figure 12 Leaderboard page, note that students emails and names were disguised.

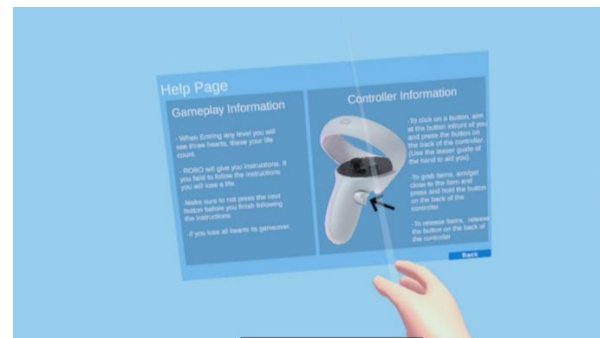


Figure 13 Help page on the controller

We have incorporated the following gamification mechanism in the application: levels, score, lives/hearts, hints, and leaderboard. We used informative sprite/icon to serve as an alert, or encouraging indicator to the user.

In terms of hardware and software requirements, the VR prototype has the following hardware requirements:

- Oculus Quest 2: Minimum Oculus Quest 2 System Requirements:
- Processor - Intel i5-4590 / AMD Ryzen 5 1500X or greater
- Graphics Card - Nvidia RTX 20 Series / AMD Radeon RX 6000 Series
- Memory - 8 GB+ RAM
- Operating System - Windows 10, Windows 11
- USB - Ports 1x USB port

The following list the software tools, IDEs, and libraries used in developing the VR prototype.

- Unity 2021.3.18f1
- Visual studio 2019
- Firebase
- Libraries and APIs used: 1XRI Interaction ToolKit and RestClient
- Meta developer account

3.3 Usability Evaluation Metrics

Usability metrics measure the system or interface's efficiency, effectiveness, satisfaction, and user-friendliness. They provide invaluable tools to objectively assess and enhance the user experience of a given application, identify areas of improvement, and validate design decisions including the value of user experience (UX) initiative of stakeholders. In this paper, we integrated both qualitative and quantitative metrics to assess our application. Table 1 presents the usability metrics we used to assess the user's experience when interacting with the VR environment. These metrics will provide objective data to support the observation and help us identify design decisions and areas of improvement.

Table 1 Usability Metrics used in this work

	Usability Quality			
Metrics	Effectiveness	Efficiency	Satisfaction	Learnability
	Task completion rate	Task time	Usability Scale	Time to first use
	Error rate	System learnability	Post-task surveys/interview	Help system usage
	Number of clicks/actions			

Effectiveness: measures how easy is for the user to complete a task. This includes the following metrics.

- **Task completion rate:** Measures the percentage of users who successfully completed a given task. This way we determine the ratio of tasks completed by task time [13].
- **Error rate:** Tracks how often users make mistakes while performing tasks.

- **Number of clicks/actions:** Measures the minimum number of steps needed to complete a task.

Efficiency: measures how quickly is for the user to accomplish a task. This includes the following metrics.

- **Task time:** Measures the average time it takes users to complete a given task. Less time can indicate efficiency but consider if speed compromises accuracy.
- **System learnability:** Measures how easily users can learn to use the software and perform basic tasks.

Satisfaction: measures how happy/satisfied are the users with their experience accomplishing the task. This includes the following metrics.

- **Usability Scale:** We developed a survey to measure the system usability based on 19 questions, see Table 2. In addition, the survey had additional essay questions for feedback and comments. We deviated from the standard System Usability Scale (SUS) because we need to emphasize more on the user interface specifics, gamification techniques, and feedback. Nonetheless, our scale includes many of the criteria in SUS. ~~This metric is a widely used questionnaire that measures users' overall subjective satisfaction with the software.~~
- **Post-task surveys:** Gather qualitative feedback about users' experiences, including likes, dislikes, and frustrations.

Learnability: measures how easy to complete basic tasks the first time it is encountered. It includes the following metrics.

- **Time to first use metric:** Measures how long it takes users to complete a basic task on their first interaction with the software.
- **Help system usage metric:** Measures how often users rely on the help system to complete tasks.

3.4 Data Sample and Collection:

Our target population is university students specialized in Information Technology, Science, and Engineering from various levels. For choosing the sample size, we are considering the following:

1. The margin of error (E) is the degree of error you can allow. In our project the margin of error that we can accept is 5%.
2. The confidence level (Z) is how sure you can be the results represent the population. The confidence level needed in our project is 90%.
3. The population size (N) is the total number of individuals from which your sample is drawn. Our population size is course section size in the UAE University which is approximately 25 students.
4. Response distribution (P) is the expected spread of results from the participants. It is how much do we expect the results to be skewed. Our response distribution is 50%.

The recommended sample size, S' , is 23 students which is calculated by Eq. 1 (Sample size for infinite population), Eq. 2. (Adjusted sample size for finite population), and using the values: $Z = 1.65$ (for 90% confidence), $P = 0.5$, $E = 0.05$, and $N = 25$ [9].

$$S = \frac{Z^2 P(1 - P)}{E^2} \quad (1)$$

$$S' = \frac{S}{1 + \frac{S - 1}{N}} \quad (2)$$

Note that using a sample of 23 students can increase feasibility by being cost-effective, time-efficient, and provide logistical ease where it is easier to manage and oversee a smaller group of participants with limited personnel or equipment. It can also enhance control and focus by allowing in-depth exploration (closer observation and richer data collection), tailored interventions, and a controlled environment. This way we can test the feasibility and effectiveness of the application before investing in larger studies, help refine the research question and hypotheses for further experimentation and allow us to identify unexpected patterns [10][11][12].

For data collection methods, this study utilized three primary data collection methods:

1. Observation Test: Participants were individually assigned a set of tasks to complete within the VR environment. A researcher observed their actions, noting task completion rates, time spent on each task, any difficulties encountered, and any user comments. This data provides objective measures of user performance and efficiency.
2. Semi-Structured Interview: Following the observation test, each participant engaged in a one-on-one interview with a researcher. The interview utilized a pre-defined set of open-ended questions designed to elicit detailed feedback on the VR application's design, functionality, and perceived ease of use. This qualitative data provides insights into user perceptions, thoughts, and suggestions for improvement, see Section 4.
3. Post-Test Survey: Upon completing the interview, participants were asked to fill out a short, anonymous survey. The survey employed a combination of Likert scale questions (e.g., strongly disagree to strongly agree) and open-ended prompts to gather further feedback on user satisfaction, usability issues, and suggestions for the VR application's development. This data allows for quantification of user sentiment and complements the qualitative insights from the interview. The survey questions included some of the SUS questions and other feedback questions. Table 2 presents the different usability criteria and the survey questions. Note that we have asked several questions on the user interface (UI) complexity and easiness because we want to identify different specifics of the UI. In addition, we asked questions on tension, colors and brightness, learnability, UI consistency, and satisfaction and engagement.

The observation test, interview questions, and responses are public and can be accessed on GitHub [16].

For the usability scale score, we performed the following steps to calculate the score for each participant.

1. For each positively-oriented question score, subtract one from the participant's answer. Answers will range from zero to nine.
2. Subtract the user responses from 10 for negatively-oriented question.
3. Sum up the adjusted score. Given the fact that we have twelve criteria, the total will range from 0 to 108.
4. Multiply the total score by 100/108 to obtain the score out of 100.

Finally, to calculate the overall usability score, we averaged the scores in step 4 over all the responses. Notice that some of the usability criteria include more than one questions. For that, we applied the same steps and averaged the score in the questions of that criteria. For example, for UI usage and complexity criteria, we average the responses in Q1, Q2, Q10, and Q15.

Table 2 The Usability scale criteria and survey questions.

Criteria	Survey Question (s)
User Interface Usage and Complexity	Q1. Did you find it easy to find the required buttons? (1 for very hard and 10 for very easy)
	Q2. Did you recognize different icons and labels? (1 for very hard and 10 for very easy)
	Q10. Were ROBO instructions clear and easy to follow?
	Q15. Was the App interface crowded and hard to use?
Memorability	Q5. Steps are easy to follow, and buttons are easy to remember (1 for no and 10 for yes)
Tension and Frustration	Q7. Was everything clear and easy to preform without frustration (1 for no and 10 for yes)?
Learnability and Technician person need	Q4. The application is easy to use with minimum instructions from the tester. (1 for no and 10 for yes)
System/interface Consistency	Q17. Is the app appearance consistent?
Performance (learning speed)	Q6. Did you manage to finish the tasks as fast as possible (1 for no and 10 for yes)?
Usage cumbersomeness.	Q8. Did you manage to perform well in labs without difficulties in using the VR equipment?
User Confidence	Q9. Did you manage to perform well in labs by hearing ROBO instructions and following them?
Prior knowledge and training	Q14. Was the navigation through the application easy and understandable?
	Q20. Could you use the application without instructions?
Brightness and Colouring	Q3. Text and colors were easy for the eyes? (1 for no and 10 for yes)
	Q13. Was the application appealing in term of colors and visuals?
Satisfaction and Engagement	Q11. You found the application interesting. (1 for no and 10 for yes)
	Q12. You found the gamification functions fun and encouraged competitive behavior. (1 for no and 10 for yes)
Recommending the App	Q23. Would you use the application in your education if available to you?
	Q24. Would you recommend the application to other people?

4. Experiments and Data Analysis

In this section, we will present and analyze the results of the observation test and the survey that we conducted during this work. First, we will present observation test results, second, we elaborate on the student survey, and finally we view the results of one-to-one interviews.

4.1 Observation Test

The usability test was conducted on 22 UAE university students from different colleges and backgrounds. The experiments spanned over 2 days for 8 hours each. The experiment flow is as follows. First, a summary of the application idea and functionality was presented to the students for 3 minutes. Second, a brief explanation of the tasks, see Table 3, to be performed by the participants was given. Third, the observer helped participants to get familiar with VR headsets and how to use the grips. Also, participants were warned about the motion sickness that the VR headset might cause, and they were instructed to inform the observer if they felt motion sickness to stop the experiment. Finally, we collected feedback and reviews through surveys and interviews. At the end of the experiment, the observers requested the student to fill in a form and they conducted an interview. It is worth mentioning that the VR Experiment's maximum time

was 30 minutes for each student and 10 minutes for the feedback. The average time to complete the maximum number of tasks of the experiment for each student was 18 minutes. In the following section, we will elaborate on the data analysis for each usability metric listed in Table 1. The technical tasks and brief description of each are shown in Table 2

Table 3 List of Tasks

Task	Assumption	Explanation
T1.1	Clicks “Register” button	The student uses the controller to click on the Register button in the landing page.
T1.2	Fill in the university/personal email and password and click Done	The student register in the application for the first time by providing viable email address and password of at least 6 characters.
T2.1	Click on the “login” button.	The student enters her user id and password to login using the keyboard.
T2.2	Fill in the newly registered email and password.	The student logs in to the application by entering her email and password.
T3.1	Proceed to the welcome page and click on the “help” icon	The user taps on the help to view topics and read/listen.
T3.2	Read the help page content and go back to the welcome page.	The help page provides an overview of the application and how to use the controllers and other tools.
T4.1	Click on the “About tools” button.	This is a simple task to check if the user can observe different icons and buttons.
T4.2	Interact with provided tools.	List of tools such as breadboard, controller, etc. are displayed and the ROBO gives information about each selected tool.
T5.1	Go back to the welcome page and click on the “labs list” button.	This is a simple task to check if the user can observe different icons and buttons.
T5.2	View labs and choose one lab to complete.	The student selects on lab to conduct and follows the ROBO instruction until completing all the steps in that lab.
T6.1	Enter the lab level and follow the instructions given by ROBO.	ROBO will give voiced instructions and highlighted hints will be provided on the breadboard.
T6.2	Users take notice if a heart is lost.	A student will have initially 3 hearts (attempts) to complete the labs. Every time the student fails to complete a task, she loses one heart.
T6.3	User loses and faces the game over page and retries.	The student will lose when she loses all 3 hearts.
T6.4	User loses and faces the game over page and goes back to lab levels list.	This is a simple task to check if the user can observe different icons and buttons.
T6.5	Click on the buttons to test their circuits.	This is a simple task to check if the user can observe different icons and buttons.
T6.6	Complete the level successfully.	The student completes all the steps in a lab (level 1 or level 2) and claims the score.
T7.1	Go back to labs list page and click on the leaderboard button.	This is a simple task to check if the user can observe different icons and buttons.
T7.2	View their score.	The student can view and see her overall score.

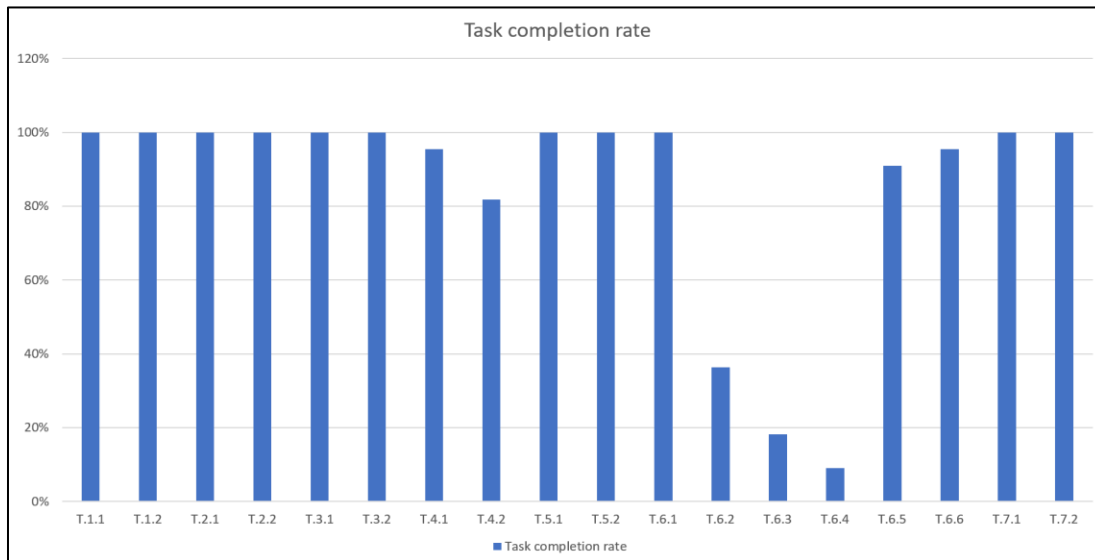


Figure 14 Task Completion Rate

Figure 14 presents the task completion rate. The x-axis of the figure represents the task as described in Table 3, and the y-axis represents the percentage of students who completed that task. We can observe that all students managed to register, log in, and at least play one level lab. We can also see that task T6.2 is skewed. Recall that Task 6.2 is relatively difficult because the student would have to move wires, gates, and connecting them to the breadboard. The majority of students have at least lost one heart (attempt) while trying the level for the first time. However, 39% of students completed all the steps in task T6.2 in the first attempt. This brings us to our next consideration: facing the game over page and retrying or leaving to lab list. 90.09% of the students did not lose all their game hearts and managed to finish the level without facing the game over page.

While the error rate during lab levels was extremely low, 31.81% of students struggled with entering their emails and passwords during registration and login. These students either wrote their email wrongly or forgot their newly registered account password, see Figure 15. The average time for the students to complete the whole list of tasks in Table 2 was 20 minutes. The task that had the longest average time was registering and logging in with an average time of 8 minutes. Table 4 compares between the real lab completion time vs. the VR lab average completion lab for the 2 level tasks. The table also presents the standard deviation and the p-value for each task. The VR lab takes less time for different reasons including the role of ROBO which assist students with online step by step instructions of how to complete the lab. Also, the logistics and setup in the physical lab take more time especially if there is only one lab assistant/instructor available for students. Additionally, the hardware/wire malfunction can be frustrating and enlarge the lab duration.

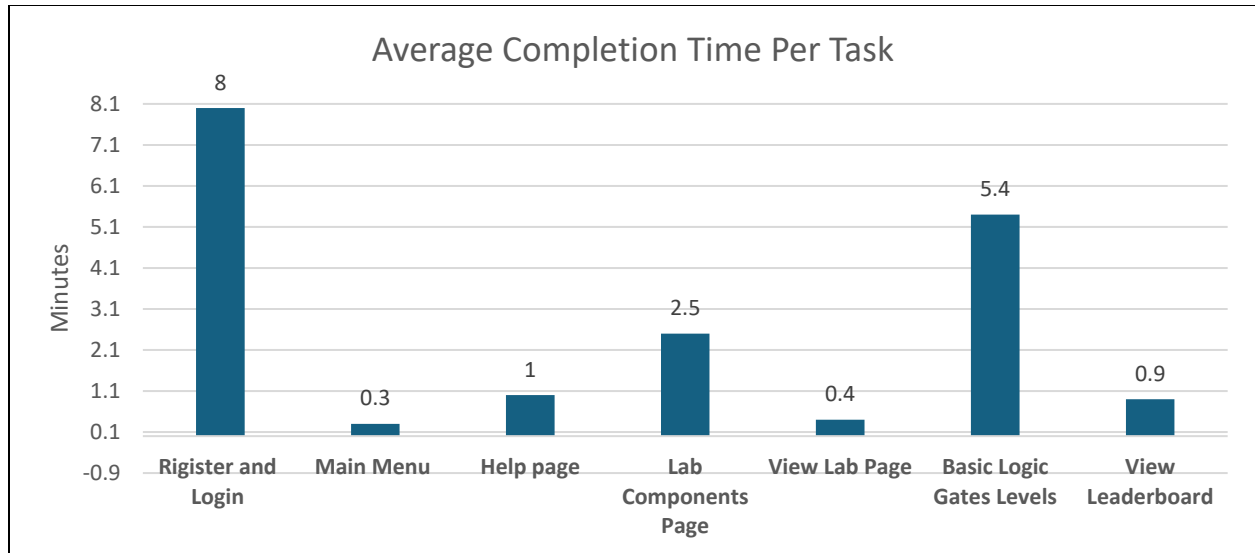


Figure 15 Average Completion Time Per Task

Table 4 Average real Lab completion vs. VR lab completion (t-test)

Basic Logic Gates T.6.6	Task Level 1		Task Level 2	
	AVG	STD	AVG	STD
Real Lab Completion Time	21 minutes	8.7 minutes	37 minutes	10.4 minutes
VR Lab Completion Time	6 minutes	2.48 minutes	10 minutes	2.8 minutes
P-Value	<0.0001		<0.0001	

While observing the students, we received the following students' comments:

1. The keyboard to log in and register needs to be closer to the user so they can interact with it more smoothly.
2. 2% of the students are nearsighted which resulted in some pictures not being clear for them in the lab levels. One student suggested putting the pictures on the VR table instead of near ROBO talk bubble to be able to work on the circuit while looking at the picture and listening to ROBO without the need to lift their head to read.
3. 9% of the students did not notice when a heart is lost and suggested to add alarming sound to make students aware of that.
4. 18% of the students showed competitiveness and played the same level multiple times to get a higher score.
5. 22.7% of students showed interest and wanted to play multiple different levels.
6. All wires have the same color (We need to make them brighter/different)

4.2 Student Survey and Usability Scale

In this section, we will look at the usability survey that we have conducted and we analyze the results and suggest improvements to the App. Questions in the survey scored on 10-point Likert scale ranging from 1 (very hard) to 10 (very easy). Additionally, we had open ended questions for the students to provide us with their feedback and comments. In the following figures, we elaborate on the main survey questions presenting the survey question, responses, average, standard deviation, and p-value. Note that the percentages of responses in the below figures are rounded. In addition, for calculating the p-value, we are using one sample test analysis with hypothetical mean of 70%.

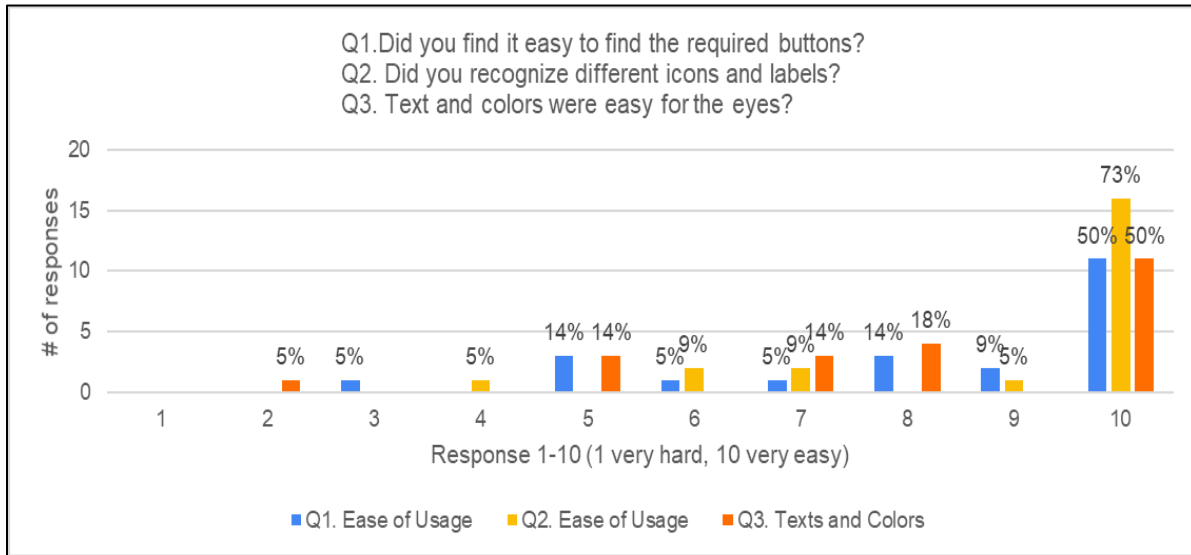


Figure 16 Responses of Q1 and Q2 (Ease of Usage) and Q3 on texts and colors.

Figure 16 presents the responses of the students in the survey when asked on the easy-of-usage of the VR application. The x-axis represents the response (1-10 where 1 is very hard and 10 is very easy) and the y-axis represents number of responses for Questions Q1-Q3. The mean, standard deviation, and the p-value of the responses are (Mean=8.31, STD=2.19, p-value=0.0089 <0.05), (Mean=9.0; STD=1.78, hypothetical mean=7, p-value<0.0001), and (Mean=8.18; STD=2.28, hypothetical mean=7, p-value=0.0212) for Q1, Q2, and Q3, respectively. The figure shows that approximately 73% of the students found the application extremely easy to find, navigate, and the text, colors, and icons were recognizable. However, 27% of the participants found it a little inconvenient and gave <8 score out of 10. After analyzing students' comments/feedback, we found that such difficulty was related to the fact that keyboard buttons and the next button were very far and hard to reach. We addressed this feedback from the student by making the keyboard closer to the student and rotate it on the x-axis slightly so it is more comfortable to use. Additionally, we made the next button closer to the table so that students can click on it as soon as they finish adding the components to the breadboard.

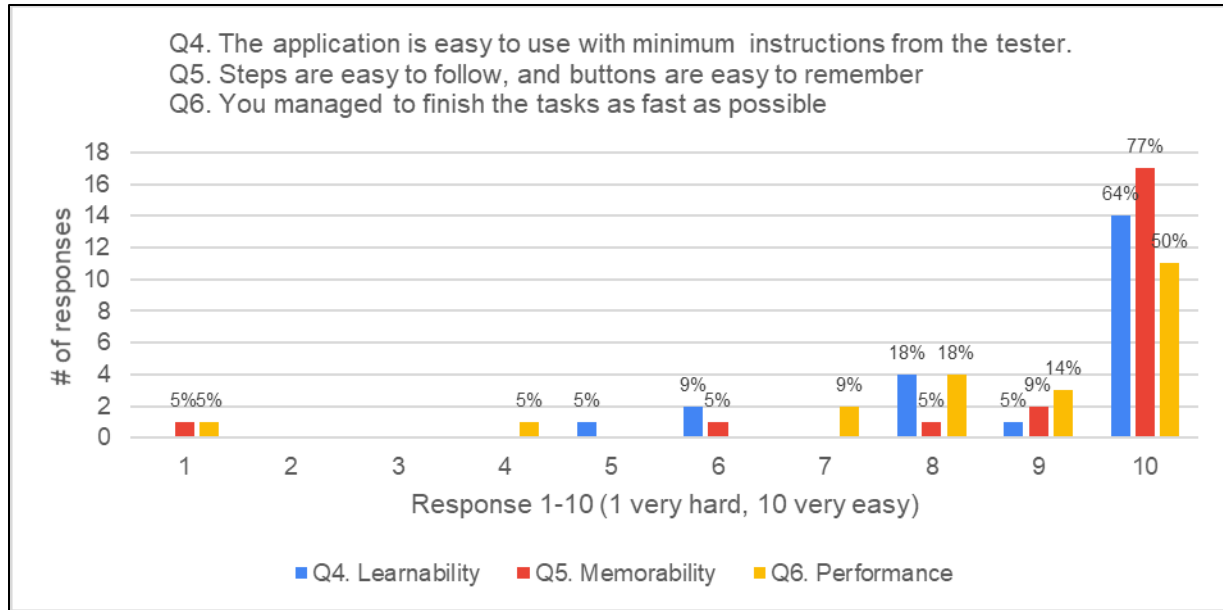


Figure 17 Responses of Q4 (Learnability), Q5 (Memorability), and Q6 (Performance)

Figure 17 presents responses on the learnability, memorability, and performance of the VR application. The figure shows that 86% of students replied that the application is easy to learn while 90% of students stated that it is easy to remember tasks and steps. The main complain of the remaining 15% of respondents was the inability to see dark color wires. Therefore, we adjusted the colors of the wires to be brighter. In terms of the performance during conducting the labs, 81% of students finished the tasks smoothly without requesting any help. The main issue of performance lagging of the remaining 19% of students was logging and registering to the application which can be mitigated by enabling other biometric methods of login such as face or fingerprint recognition. The mean, standard deviation, and the p-value of the responses are (Mean=9; STD=1.57, p-value<0.0001), (Mean=9.22; STD=2.06, p-value<0.0001), and (Mean=8.54, STD=2.26, p-value=0.0035) for Q4, Q5, and Q6, respectively.

Figure 18 presents the responses of participants in terms of tension/frustration, usage of VR equipment, and Robo instructions. We can see that 77% of the students did not feel any tension or frustration while 33% of students stated that they need more practice to get better experience with the VR controls and equipment. We obtained similar results on the use of VR headset and equipment in the application. In addition, 90% of the students managed to complete lab levels with only ROBO instructions. The mean, standard deviation, and the p-value of the responses are (Mean=8.5, STD=2.57, p-value=0.0105), (Mean=8.72, STD=2.33, p-value=0.0018), and (Mean=9.31; STD=2.0, p-value<0.0001) for Q7, Q8, and Q9, respectively.

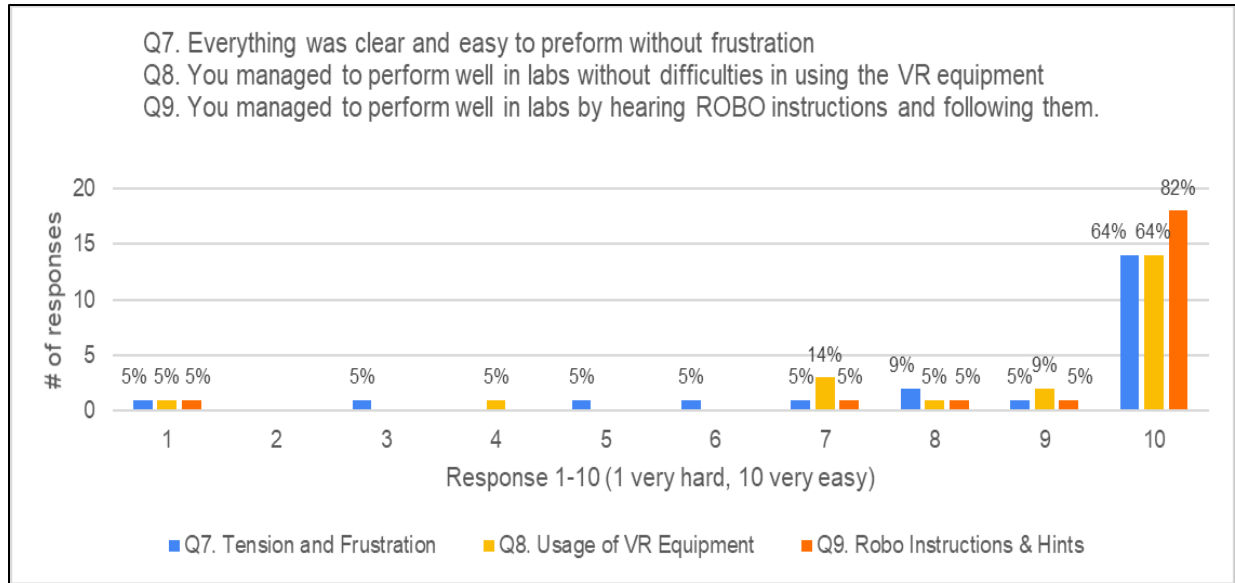


Figure 18 Responses of Q7 (Tension and Frustration), Q8 (Usage of VR equipment), and Q9 (Robo Instructions and hints)

They thought that the hints and voiced instruction of ROBO were very helpful and easy to follow. Additionally, the Figure 19 shows that 95% of students are satisfied with the application and with the gamification elements.

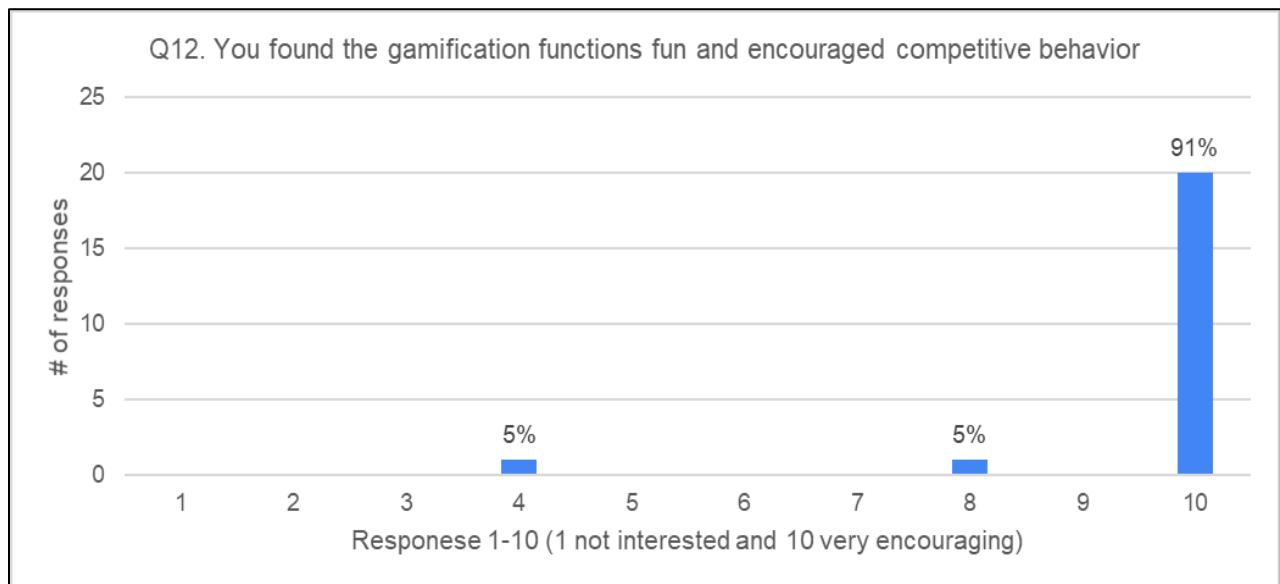


Figure 19 Satisfaction: students' interest and the effect of gamification (Mean=9.63; STD=1.32, hypothetical mean=7, p -value<0.0001).

Table 5 summarizes the survey results and presents the average and standard deviation for the 12 usability criteria. The overall usability scale we obtained is 84% which indicates that the system's usability is excellent.

Table 5 Usability Survey Questions results (average and Standard deviation)

Criteria	Survey Question (s)	AVG	STD
User Interface Usage and Complexity	Q1, Q2, Q10, and Q15	86.87	14.75
Memorability	Q5	91.41	22.98
Tension and Frustration	Q7	83.33	28.64
Learnability & technician person need	Q4	88.89	17.48
System/interface Consistency	Q17	71.72	20.5
Performance (Learning Speed)	Q6	83.84	25.13
Usage cumbersomeness.	Q8	85.86	25.93
User Confidence	Q9	92.42	22.32
Prior knowledge and training	Q14, Q20	71.72	20.21
Brightness and Colouring	Q3, Q13	76.77	13.89
Satisfaction and Engagement	Q11, A12	97.22	8
Recommending the App	Q23, Q24	78.79	22.46
Overall Usability		84.06%	11.85

4.3 One-to-One Student Interview

In addition to the survey, we conducted one-to-one interview with the students. In the following we highlight the main issues and remarks from the students.

- 90% of the students thought that the VR application was not crowded, easy to use, close to the real lab, simple, clear, and well organized.
- 95% of the students thought the app's appearance was consistent and not confusing.
- 86.4% of the students thought that they could use the application without instructions.

Additionally, we received the following remarks on the fun points that enhanced students experience in the application:

- “Putting all the chips and wires together.”
- “Starting the circuit”, after completing the lab.
- “To get the result that I was expecting from the lab.”
- “I liked experimenting with the wires and doing the labs and I loved the leaderboard.”
- “The limited hearts made me think twice about the placement, so I don’t lose, and it’s fun to compete with your colleagues in the leaderboard.”
- “The physical board with wires and other components.”
- “Switching the LED ON and OFF”
- “The part where it showed me that I was wrong in some of the applications, and I was very satisfied when it worked.”
- “Leaderboard time and hearts.”

- “Performing the task was easy and well-guided for someone who is playing for the first time.”
- “The way we can interact to build the circuit with instructions.”
- Introduce “Change Language” option and have Arabic/English languages.

5. Discussion

The usability study of the Digital Design & Computer Organization lab VR application yielded valuable insights into user experience within the immersive VR environment. By analyzing the data collected, this section will discuss the key findings and their implications for improving the application's effectiveness.

Generally, the data analyzed in the previous section showed that majority of students highly appreciated the user experience with the application. The usability test identified several areas for improvement in the application. Therefore we provided an updated version of the application with additional fixes and features. For instance, users reported issues differentiating wire colors. After the update, wires' colors become brighter and visible/distinguishable. Another issue was overlapping voice lines in the lab components page, the ability to choose two tools at the same time which led to confusion, and some users thought that some of the text size was too small. All the lab components page mentioned issues were fixed and updated. Another issue was viewing the 1bit comparator leaderboard would sometimes not load. This issue was fixed as well. Furthermore, the following list shows other general updates:

- Added soundtrack (BGM, heart lost sound effect, game over sound effect).
- New hint added to levels: “My notes paper” contains lab tools used.
- Changed the placement of logic gates diagrams in levels to be on the table to make it easier for the student to follow instructions.
- New hint added to levels: Add VCC and GND text on the board.
- Fixing some register/login issues such as requesting only the university instead of their full email.
- General fixes.

In terms of engineering education, the study found the following:

- Learnability: The VR Lab was easy to learn with minimum interaction. (Mean=9; STD=1.57, hypothetical mean=7, p-value<0.0001)
- Memorability: the VR application makes significant improvement for students to easily remember steps (Mean=9.22; STD=2.06, hypothetical mean=7, p-value<0.0001).
- The VR Lab enabled the students to finish tasks quickly (Mean=8.54; STD=2.26, hypothetical mean=7, p-value=0.0035).
- The VR Lab reduced the task completion time; hence, it reduced the amount of tension and frustration that might result in hardware failure and lack of resources and guidance (Mean=8.5; STD=2.57, hypothetical mean=7, p-value=0.0105).

Compared to traditional setting of the Lab, the above findings have great motivation to integrate the VR-Lab in the course. In addition, the feedback of the students showed that such VR-Lab contributed positively on their education and engagement during the lab.

Finally, we found during this work that the use of VR application in education necessitates conducting thorough usability testing before deployment. The usability test is a paramount to identify weaknesses and bugs in the application. We found that gamification is very effective in education to encourage continuation, competitive behavior, satisfaction, and great user experience.

6. Limitations of the study

In this paper, we investigated the students' preferences in the VR application to learn digital design course labs. We analyzed the survey results on the student's experience with the VR environment and their acceptance of the technology. Students from different colleges applied for the experiment, however, the sample size of 22 students may not be generalizable to the entire population of the Digital Design & Computer Organization lab course. Also, the study focused on the Digital Design & Computer Organization lab VR environment limiting the findings to only this educational course rather than other courses that can be implemented in VR. Additionally, there was a shortage of resources, we had only one VR headset to test with, thus the experiment was time-consuming. Future research with larger and more diverse participant groups, as well as studies conducted in actual classrooms, could provide more comprehensive insights.

7. Ethical Considerations

The surveys in this work were conducted according to the guidelines Institutional Review Board (or Ethics Committee) of the United Arab Emirates University (protocol code ERSC_2022_2029 and date of approval 23/02/2023), copy of the ethical form can be found at [16]. Informed consent was obtained from all subjects involved in the study. Prior to starting of the survey and the experiment, the authors explained to the students the procedure and the usage of the application. The purpose of the study was not explained in advance so as not to influence students' responses. Students were instructed that they can discontinue participation or decline to respond to any of the questions in the survey. All students provided spoken informed consent.

8. Conclusion

In conclusion, this usability study discussed the user experience of the Digital Design & Computer Organization lab VR application. By employing a mixed-methods approach with observation tests, interviews, and surveys, the study yielded valuable insights into user performance and perceptions.

Overall, user feedback indicated positive perceptions of the VR application's ability to gamify the Digital Design & Computer Organization lab experience. Participants found that the immersive environment engaging, competitive, and helpful for understanding the material. However, some users reported challenges with registering and logging in. This study also emphasizes the importance of iterative design and user testing throughout the development process of VR applications in education.

The usability study has shown that VR-Lab contributed positively on the students' engagement and interest in the lab materials. In addition, the VR-Lab can lower the level of frustration and tension during the lab. Therefore, we recommend to integrate the VR-Lab in the course to support students' learning and to make such engineering complex concepts fun to learn. However, the use of physical breadboard and wires will continue due to engineering and course requirements.

For future research, we plan to deploy the VR application in a live course and investigate the impact of gamification and VR on the academic level of students and the learning outcomes achievement of the course. We also plan to study the impact of VR-Lab on physical task completion time.

References

1. Al-Ansi, Abdullah M., *et al.* "Analyzing augmented reality (AR) and virtual reality (VR) recent development in education." *Social Sciences & Humanities Open* 8.1 (2023): 100532.
2. Akcayir, M., & Akcayir, G. (2017). Advantages and challenges of virtual reality in education. *International Journal of Virtual and Reality Education*, 6(2), 1-12.
3. Kamińska, D., Zwoliński, G., & Laska-Leśniewicz, A. (2022). Usability testing of virtual reality applications—the pilot study. *Sensors*, 22(4), 1342.

4. Hartson H.R., Andre T.S., Williges R.C. Criteria for evaluating usability evaluation methods. *Int. J. Hum. Comput. Interact.* 2001;13:373–410. doi: 10.1207/S15327590IJHC1304_03
5. Chandra A.N.R., El Jamiy F., Reza H. A review on usability and performance evaluate on in virtual reality systems; Proceedings of the 2019 International Conference on Computational Science and Computational Intelligence (CSCI); Las Vegas, NV, USA. 5 December 2019; pp. 1107–1114.
6. Brade, J., Lorenz, M., Busch, M., Hammer, N., Tscheligi, M., & Klimant, P. (2017). Being there again – Presence in real and virtual environments and its relation to usability and user experience using a mobile navigation task. *International Journal of Human-computer Studies*, 101, 76–87.
7. Lucas J.D. Immersive VR in the construction classroom to increase student understanding of sequence, assembly, and space of wood frame construction. *J. Inf. Technol. Constr.* 2018;23:179–194
8. Wolfartsberger J. Analyzing the potential of Virtual Reality for engineering design review. *Autom. Constr.* 2019;104:27–37. doi: 10.1016/j.autcon.2019.03.018
9. Raosoft Inc. (2004) RaoSoft® sample size calculator. <http://www.raosoft.com/samplesize.html>. Accessed 06 February 2024.
10. Field, A. P. (2009). *Discovering statistics using SPSS* (3rd ed.). SAGE Publications Ltd. (Chapter 5 discusses sample size considerations and trade-offs)
11. Machin, V. D., & Swiss, P. (2021). *Practical Statistics for Economics and Finance* (5th ed.). Routledge. (Chapter 3 covers sample size determination and power analysis)
12. Maxwell, S. E., & Cole, D. A. (2007). *Research Methods in Education* (2nd ed.). Wadsworth Publishing.) (Chapter 4 discusses sample size considerations and justifications)
13. Sampaio, A. M., & Barbosa, J. G. (2016). Energy-Efficient and SLA-Based resource management in cloud data centers. In *Advances in Computers* (pp. 103–159).
14. Kurbatova, S. M., L. Yu Aisner, and S. D. Krasnousov. "Virtual Technologies in the Educational Space: Pros and Cons." (2022.)
15. Gawron, Valerie J. "Review of the effects of fatigue on performance of laboratory tasks." *International Journal of Human Factors and Ergonomics* 3.1 (2014): 86-103.
16. GitHub Link to Supportive Documents and Article Resources. Available online: <https://github.com/MariamA-A/Digital-Design-Computer-Organization-VR-Lab> (accessed on May 24, 2024)
17. Dias, Paulo, *et al.* "Usability in Virtual and Augmented Environments: A qualitative and quantitative study." *Stereoscopic Displays and Virtual Reality Systems XIV*. Vol. 6490. SPIE, 2007.
18. Huang, H., & Lee, C. F. (2022). Factors affecting usability of 3D model learning in a virtual reality environment. *Interactive Learning Environments*, 30(5), 848-861.
19. 8 Soliman, M., Pesyridis, A., Dalaymani-Zad, D., Gronfula, M., & Kourmpetis, M. (2021). The application of virtual reality in engineering education. *Applied Sciences*, 11(6), 2879.
20. 9 di Lanzo, J. A., Valentine, A., Sohel, F., Yapp, A. Y., Muparadzi, K. C., & Abdelmalek, M. (2020). A review of the uses of virtual reality in engineering education. *Computer Applications in Engineering Education*, 28(3), 748-763.
21. 10 Huang, W., & Roscoe, R. D. (2021). Head-mounted display-based virtual reality systems in engineering education: A review of recent research. *Computer Applications in Engineering Education*, 29(5), 1420-1435.
22. 12 Oje, A. V., Hunsu, N. J., & May, D. (2023). Virtual reality assisted engineering education: A multimedia learning perspective. *Computers & Education: X Reality*, 3, 100033.
23. Feisel, L. D., & Rosa, A. J. (2005). The role of the laboratory in undergraduate engineering education. *Journal of engineering Education*, 94(1), 121-130.

24. Kollöffel, B., & De Jong, T. (2013). Conceptual understanding of electrical circuits in secondary vocational engineering education: Combining traditional instruction with inquiry learning in a virtual lab. *Journal of engineering education*, 102(3), 375-393.
25. Maroungas, A., Troussas, C., Krouska, A., & Sgouropoulou, C. (2023). Virtual reality in education: a review of learning theories, approaches and methodologies for the last decade. *Electronics*, 12(13), 2832.
26. Jochecová, K.; Černý, M.; Stachon, Z.; Švedová, H.; Káčová, N.; Chmelík, J.; Bružá, V.; Kvarda, O.; Ugwitz, P.; Šašinková, A.; *et al.* Geography Education in a Collaborative Virtual Environment: A Qualitative Study on Geography Teachers. *ISPRS Int. J. Geo-Inf.* 2022, 11, 180.
27. Wee, C., Yap, K. M., & Lim, W. N. (2022). iprogvr: Design of a virtual reality environment to improve introductory programming learning. *IEEE Access*, 10, 100054-100078.
28. Tarnag, W., Chen, C. J., Lee, C. Y., Lin, C. M., & Lin, Y. J. (2019). Application of virtual reality for learning the material properties of shape memory alloys. *Applied Sciences*, 9(3), 580.
29. Al Kork, S., & Beyrouthy, T. (2018). Interactive virtual reality educational application. *Adv. Sci. Technol. Eng. Syst. J*, 3, 72-82.
30. Freina, L., & Ott, M. (2015, April). A literature review on immersive virtual reality in education: state of the art and perspectives. In *The international scientific conference elearning and software for education* (Vol. 1, No. 133, pp. 10-1007).
31. Li, Z., Wang, W., & Huang, Z. Z. (2025). Virtual reality for emergency medicine training on traffic accident injury treatment: a randomized controlled trial. *Frontiers in Virtual Reality*, 6, 1518016.
32. Birrenbach, T., Stuber, R., Müller, C. E., Sutter, P. M., Hautz, W. E., Exadaktylos, A. K., ... & Sauter, T. C. (2024). Virtual reality simulation to enhance advanced trauma life support trainings—a randomized controlled trial. *BMC medical education*, 24(1), 666.
33. Jung, Y. (2022). Virtual reality simulation for disaster preparedness training in hospitals: integrated review. *Journal of Medical Internet Research*, 24(1), e30600.
34. Scorgie, D., Feng, Z., Paes, D., Parisi, F., Yiu, T. W., & Lovreglio, R. (2024). Virtual reality for safety training: A systematic literature review and meta-analysis. *Safety science*, 171, 106372.
35. Deriba, F. G., Saqr, M., & Tukiainen, M. (2024, June). Assessment of accessibility in virtual laboratories: a systematic review. In *Frontiers in Education* (Vol. 9, p. 1351711). Frontiers Media SA.
36. Norwine, L., Bouck, E., Reiley, S., Long, H. M., Jakubow, L., & Nuse, J. (2025). Using Non-Immersive Virtual Reality to Teach Life Skills to Students With Disabilities in Rural Schools. *Rural Special Education Quarterly*, 44(1), 26-38. <https://doi.org/10.1177/87568705241305640> (Original work published 2025)
37. Eaves, D. L., Breslin, G., Van Schaik, P., Robinson, E., & Spears, I. R. (2011). The short-term effects of real-time virtual reality feedback on motor learning in dance. *Presence: Teleoperators and Virtual Environments*, 20(1), 62-77.
38. Shi, L., & Xu, C. (2024). Virtual reality technologies to provide performance feedback for motor and imagery training. *Education and Information Technologies*, 1-19.
39. Farra, S. L., Gneuhs, M., Hodgson, E., Kawosa, B., Miller, E. T., Simon, A., ... & Hausfeld, J. (2019). Comparative cost of virtual reality training and live exercises for training hospital workers for evacuation. *CIN: Computers, Informatics, Nursing*, 37(9), 446-454.
40. Veras, M., Stewart, J., Deonandan, R., Tatmatsu-Rocha, J. C., Higgins, J., Poissant, L., & Kairy, D. (2020). Cost analysis of a home-based virtual reality rehabilitation to improve upper limb function in stroke survivors. *Glob J Health Sci*, 12(2), 98-105.
41. Ironsi, C. S. (2023). Investigating the use of virtual reality to improve speaking skills: insights from students and teachers. *Smart Learning Environments*, 10(1), 53.

42. Hernández-de-Menéndez, M., Vallejo Guevara, A., & Morales-Menendez, R. (2019). Virtual reality laboratories: a review of experiences. *International Journal on Interactive Design and Manufacturing (IJIDeM)*, 13, 947-966.
43. Alnagrat, A. J. A., Ismail, R. C., & Idrus, S. Z. S. (2022). The Opportunities and challenges in virtual reality for virtual laboratories. *Innovative Teaching and Learning Journal*, 6(2), 73-89.
44. Lynch, T., & Ghergulescu, I. (2017). Review of virtual labs as the emerging technologies for teaching STEM subjects. *INTED2017 proceedings*, 6082-6091.
45. Hoorn, J. F., Konijn, E. A., & Van der Veer, G. C. (2003). Virtual reality: Do not augment realism, augment relevance. *Human-Computer Interaction: Overcoming Barriers*, 4(1), 18-26.
<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=e4090d5f5e5b87ab6a2dc1d88fb22c5e0c20f025>
46. Newman, M. A. R. K., Gatersleben, B., Wyles, K. J., & Ratcliffe, E. (2022). The use of virtual reality in environment experiences and the importance of realism. *Journal of environmental psychology*, 79, 101733.
47. Jung, S., & Lindeman, R. W. (2021). Perspective: Does realism improve presence in VR? suggesting a model and metric for VR experience evaluation. *Frontiers in virtual reality*, 2, 693327.
48. Romli, R., & Yaakob, A. R. (2017). The theoretical model of designing desktopVR in learning environment. *Advanced Science Letters*, 23(5), 4211-4214.
49. Gonçalves, G., Coelho, H., Monteiro, P., Melo, M., & Bessa, M. (2022). Systematic review of comparative studies of the impact of realism in immersive virtual experiences. *ACM Computing Surveys*, 55(6), 1-36.
50. Slater, M., & Wilbur, S. (1997). A framework for immersive virtual environments (FIVE): Speculations on the role of presence in virtual environments. *Presence: Teleoperators & Virtual Environments*, 6(6), 603-616.
51. Katernyak, I., & Loboda, V. (2016). Cognitive Presence and Effect of Immersion in Virtual Learning Environment. *Universal Journal of Educational Research*, 4(11), 2568-2573.
52. Selzer, M. N., Gazcon, N. F., & Larrea, M. L. (2019). Effects of virtual presence and learning outcome using low-end virtual reality systems. *Displays*, 59, 9-15.
53. Lu, F., Yu, D., Liang, H. N., Chen, W., Papangelis, K., & Ali, N. M. (2018, October). Evaluating engagement level and analytical support of interactive visualizations in virtual reality environments. In *2018 IEEE international symposium on mixed and augmented reality (ISMAR)* (pp. 143-152). IEEE.
54. Zhang, L., Bowman, D. A., & Jones, C. N. (2019, September). Exploring effects of interactivity on learning with interactive storytelling in immersive virtual reality. In *2019 11th international conference on virtual worlds and games for serious applications (VS-games)* (pp. 1-8). IEEE.
55. de França, A. C. P., Pereira Neto, J., & Soares, M. M. (2018). Methods and procedures to usability testing in virtual reality systems. In *Advances in Ergonomics in Design: Proceedings of the AHFE 2017 International Conference on Ergonomics in Design*, July 17– 21, 2017, The Westin Bonaventure Hotel, Los Angeles, California, USA 8 (pp. 45-51). Springer International Publishing.
56. Chen, C. J., Lau, S. Y., Chuah, K. M., & Teh, C. S. (2013). Group usability testing of virtual reality-based learning environments: A modified approach. *Procedia-Social and Behavioral Sciences*, 97, 691-699.
57. Downey, L. L. (2007). Group usability testing: Evolution in usability techniques. *Journal of Usability Studies*, 2(3), 133-144.
58. Zilles Borba, E., Corrêa, A. G., de Deus Lopes, R., & Zuffo, M. (2020). Usability in virtual reality: evaluating user experience with interactive archaeometry tools in digital simulations. *Multimedia tools and applications*, 79(5), 3425-3447.

59. Chisalita, C., Puerta-Melguizo, M. C., & Van der Veer, G. C. (2003). Designing Complex Systems in Industrial Reality: A Study of the DUTCH Approach. An accepted European ICT certification standard promoted by CEPIS (Council of European Professional Informatics Societies), 46.